Cosmin Suna
cosminsuna@yahoo.com | (514) 961-8065 | LinkedIn

Academic Background

Concordia University – Bachelor of Software Engineering, Co-op Program | 2019 – 2025
John Abbott College – DEC, Pure and Applied Sciences | 2017 – 2019
Saint-Louis College – DES & International Baccalaureate | 2012 – 2017

Skills Summary

Programming Languages: C#, Java, Python, C++, C, HTML, CSS, JavaScript (React), Erlang, Lisp, Prolog
Databases / OS: MySQL, PostgreSQL, SQL Server (ADO.NET), Windows, Linux
Tools / Engines: Unity, Blender, Aseprite, Git/GitHub, VS/VS Code, Node.js, Flask, Bootstrap, Microsoft Suite
Soft Skills: Collaboration & teamwork, problem-solving & adaptability, analytical & detail-oriented
Spoken Languages: English, French, Romanian

Work Experience

Neptronic – Full-Stack Web Programmer Intern | 01/2021 – 04/2021

- Developed and maintained **enterprise ASP.NET MVC modules** supporting sales, order processing, production, and supply chain management.
- Designed **responsive user interfaces** and integrated dynamic content; created reusable **React and .NET components** to enhance usability.
- Implemented a **secure credential storage system** using PBKDF2 with salt + hash, migrating legacy plaintext passwords to **industry-standard encryption**.
- Built **backend utilities in C#** to generate salts, hash passwords, verify credentials, and update **User/WebUser tables** across multiple production databases.
- Wrote **optimized SQL queries and update scripts** for authentication workflows, including parameterized statements to prevent **SQL injection vulnerabilities**.
- Automated **full-table password migrations** (4,000+ accounts) by retrieving records, generating salted hashes, and performing **batch updates**.
- Troubleshot authentication and database issues, enhancing **data integrity**, system reliability, and compatibility with legacy modules.
- Managed **SQL Server connections and data readers** using ADO.NET, optimizing query execution and reducing runtime errors.
- Conducted software testing, identified **technical defects**, and improved **database operation stability**.

Next-Generation Cities Institute – Unity Developer | 09/2024 – 05/2025

- Collaborated with a 10-member team to develop an interactive urban planning simulation for evaluating street walkability.
- Implemented core simulation parameters in Unity, including intersection generation, greenery placement, and building enclosure algorithms.
- Designed and integrated real-time environmental customization features, allowing users to adjust greenery, connectivity, public spaces, and enclosure during first-person navigation.

- Optimized assets and scripts to improve simulation performance and reduce runtime overhead in large urban environments.

Key Projects

Movie Analytics with SQL

- Designed and implemented a relational database in PostgreSQL for a large real-world dataset (0.5GB+ across multiple files).
- Built a full **ER model** and translated it into SQL **DDL** to create optimized relational schemas.
- Loaded and cleaned multi-file datasets into PostgreSQL using bulk import tools.
- Developed **10+ analytical SQL reports** using joins, aggregate functions, GROUP BY, ORDER BY, and filtering.
- Created and tested **indexes** to improve query performance and measure execution speedups.
- Compared SQL results with a NoSQL system by modeling the same dataset in a non-relational format (e.g., Elasticsearch/Neo4J/etc.).

Flight Tracking System

- Collaborated in a 3-person team to design and prototype an object-oriented **flight tracking system** similar to FlightRadar24.
- Produced full **software requirements artifacts**, including use cases, UML domain model, system sequence diagrams, operation contracts, interaction diagrams, and class diagrams.
- Designed system behavior for multiple actors—system admin, airport admin, airline admin, and end-users—with role-based access control.
- Modeled concurrency rules (reader/writer mutual exclusion) and ensured consistency across all system artifacts.
- Implemented core use cases: viewing flights, registering flights, and maintaining persistent records.
- Designed a **relational data model** and integrated database persistence for flights, airports, aircraft, airlines, and cities.
- Delivered a functional prototype and a final **demonstration video** showcasing system behavior and end-to-end workflow.
- Used GitHub for version control, task coordination, and individual workload tracking

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Personal Achievements / Projects

- Selected to represent Concordia University at **Ubisoft Gamelab 2025**.
- Earned Ubisoft Game Creator's Odyssey certificates: Rational Game Design & Rational Level Design.
- Released **AstroDrop** on Steam; developed additional indie games including AI-driven strategy simulations and casual mobile-style prototypes (tank-themed Temple Run, match-3).

Portfolio Links

- Personal Website
- coscos100.itch.io